



A Cross-Species Lexical Repertoire Reveals a Biological Code Linking Acoustic Form to Communicative Meaning

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Abstract

Understanding animal vocalizations means connecting acoustic form to communicative meaning. Years of field studies have produced extensive knowledge of such mappings, yet no unified framework exists for comparing them across species. We introduce AnimalLex, a cross-species repertoire database comprising 1,507 calls from 128 species spanning mammals, birds, and amphibians, annotated with standardized semantic and acoustic descriptions synthesized from the scientific literature. Using AnimalLex, we characterize what animals communicate about and how. Acoustic and semantic distances are significantly correlated within species, across related species, and across families, providing evidence for form-to-meaning structure that extends across phylogenetic boundaries. We leverage this structure to identify cross-species acoustic–semantic motifs and to support lexical exploration between repertoires. These results suggest that a biological code linking acoustic form to communicative function may be a broad organizational principle of animal communication, and that large-scale repertoire synthesis—now made practical by language models—unlocks new possibilities for comparative bioacoustics. An interactive tool provides the complete results and supports exploration at scale.

1 Introduction

Decades of field research have established that animal vocalizations are not simple reflexes but structured communicative signals whose forms are shaped by function [Marler, 1955](#). Alarm calls are short, broadband, and easily localized—suited for rapid threat broadcast. Contact calls are tonal and repetitive—suited for long-distance propagation and individual recognition. Infant-directed calls are high-pitched across taxa—plausibly linked to shared attentional mechanisms. Yet these patterns have been documented one species at a time, and no synthesis has asked whether they constitute a universal code: a systematic, learnable relationship between acoustic form and communicative meaning that holds across the tree of life.

The question is not merely empirical but conceptual. In linguistics, the arbitrariness of the sign—the principle that a word’s sound bears no necessary relation to its meaning—has been a foundational tenet since Saussure [Saussure, 1916](#). Counter-evidence is accumulating: sound symbolism and crosslinguistic iconicity demonstrate that form-to-meaning mappings exist and are learned in human language [Dingemanse et al., 2015](#). In non-human animals the question is more open still. If such a biological code exists, it could arise through two mechanisms: convergent evolution driven by shared physical and ecological constraints (e.g., the localizability demands of alarm signaling), or phylogenetic conservation of ancestral form-to-meaning associations within lineages. These mechanisms make different predictions—the former produces cross-taxon universals, the latter family-level regularities—and can in principle be distinguished with sufficient data.

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Here we test for a biological code at scale by assembling AnimalLex, a database of 1,507 calls from 128 species (Fig. 1). Each entry pairs a free-text acoustic description with a free-text semantic description, standardized ontology keywords, confidence, and references synthesized from literature search. We embed both descriptions independently into continuous vector spaces and ask whether their pairwise distance structures are aligned—within species, across species, and across higher taxa. A simple linear model trained on this alignment is evaluated on held-out calls, species, and families. We use the resulting spaces to identify cross-species acoustic–semantic motifs and provide an interactive web application for broad exploration.

2 A cross-species lexical repertoire

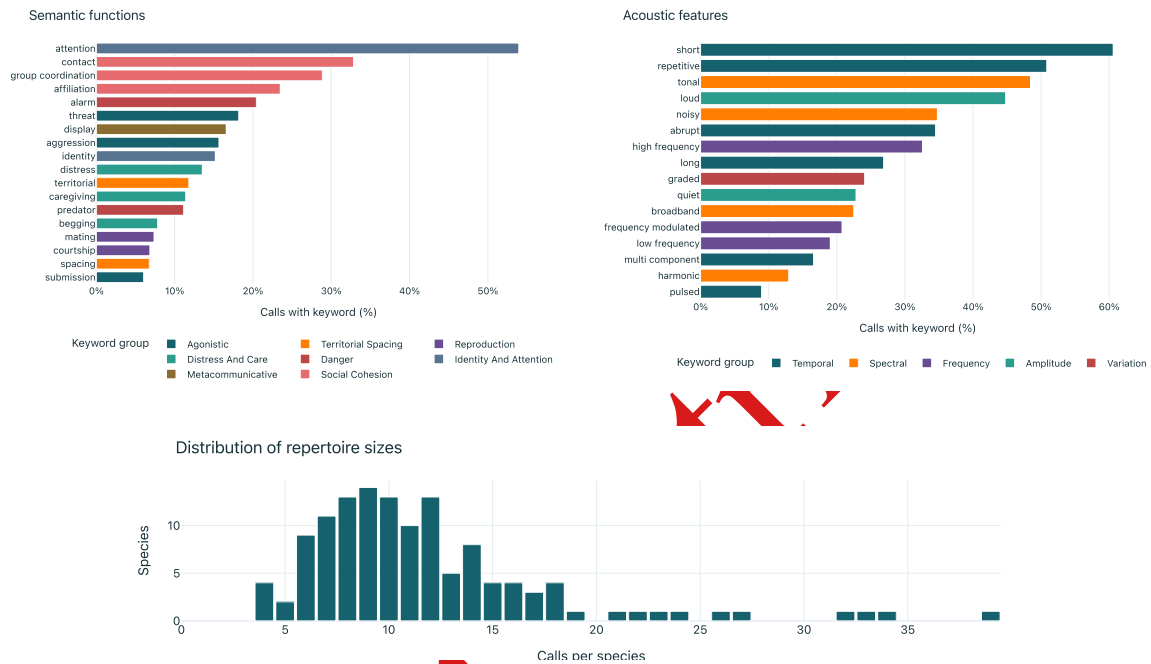


Figure 1. AnimalLex: a curated cross-species lexical repertoire. (A) The 18 most frequent semantic ontology keywords across all 1,507 calls, grouped by communicative function. (B) The 18 most frequent acoustic features across the same calls, grouped by feature type. (C) Distribution of documented repertoire sizes across 128 species. All panels are static exports of the figures displayed in the web application’s unfiltered view.

AnimalLex was assembled by prompting a large language model to structure knowledge from peer-reviewed literature for each target species, producing entries that record call name, free-text acoustic description, free-text semantic description, standardized ontology keywords, confidence, and references (see Methods). The 128 species span three vertebrate classes: 82 mammals, 37 birds, and 9 amphibians, covering taxa from African savanna elephants (*Loxodonta africana*) and gray wolves (*Canis lupus*) to Carolina chickadees (*Poecile carolinensis*) and túngara frogs (*Engystomops pustulosus*).

The most common semantic keywords are attention ($n = 813$), contact ($n = 495$), group coordination ($n = 435$), affiliation ($n = 354$), and alarm ($n = 308$; Fig. 1A). The most frequent acoustic features are short ($n = 913$), repetitive ($n = 766$), and tonal ($n = 730$; Fig. 1B). Together these annotation axes provide the substrate for testing form-to-meaning alignment.

3 Acoustic and semantic spaces are aligned across the tree of life

We embedded each call’s acoustic and semantic descriptions separately into a shared sentence-transformer space (see Methods) and computed pairwise cosine similarities across all 1,134,771 call pairs. Blocked Mantel tests (Mantel, 1967) show positive acoustic–semantic alignment within species ($r = 0.16$, $p = 0.001$, $n = 10,452$ pairs), among different species in the same family ($r = 0.19$, $p = 0.001$, $n = 40,977$), and among species in different families ($r = 0.14$, $p = 0.001$, $n = 1,083,342$; Fig. 2).

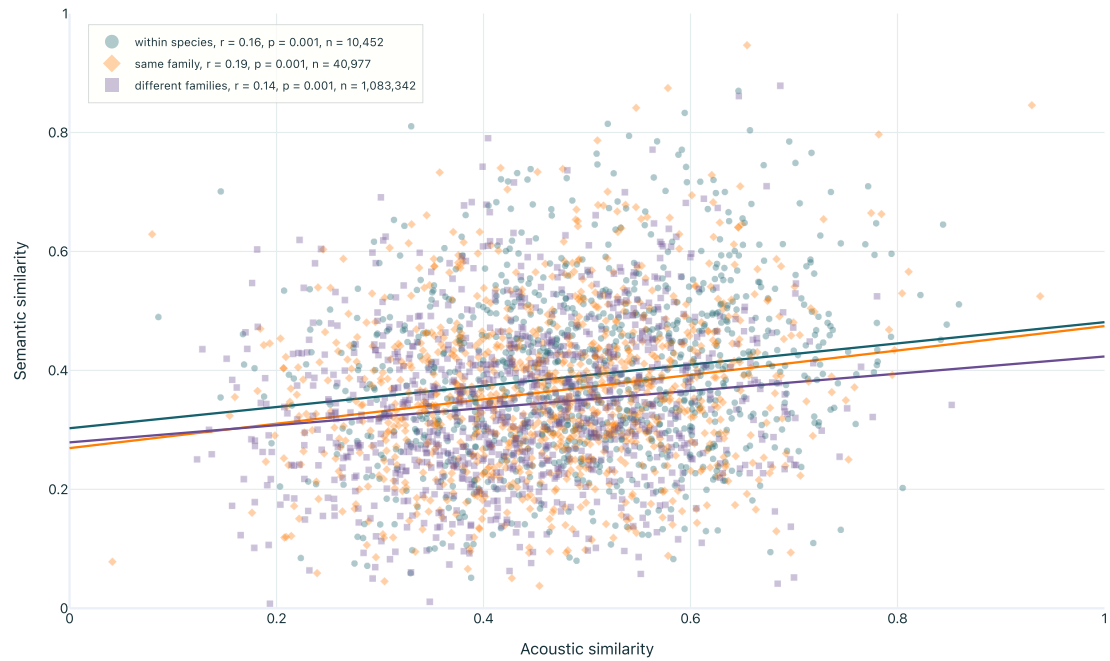


Figure 2. Alignment of acoustic and semantic embedding spaces. Acoustic similarity versus semantic similarity for the samples displayed in the web application, separated into within-species, same-family cross-species, and different-family pairs. Lines show the corresponding linear trends; legends report Mantel r , permutation p , and the number of pairs in the complete analysis.

The same-family coefficient is the strongest of the three, although the difference is modest. This pattern is compatible with a contribution from shared evolutionary history, while the positive different-family coefficient shows that the association is not restricted to close relatives. Figure 3 shows the species-pairwise correlations underlying the web application's matrix view.

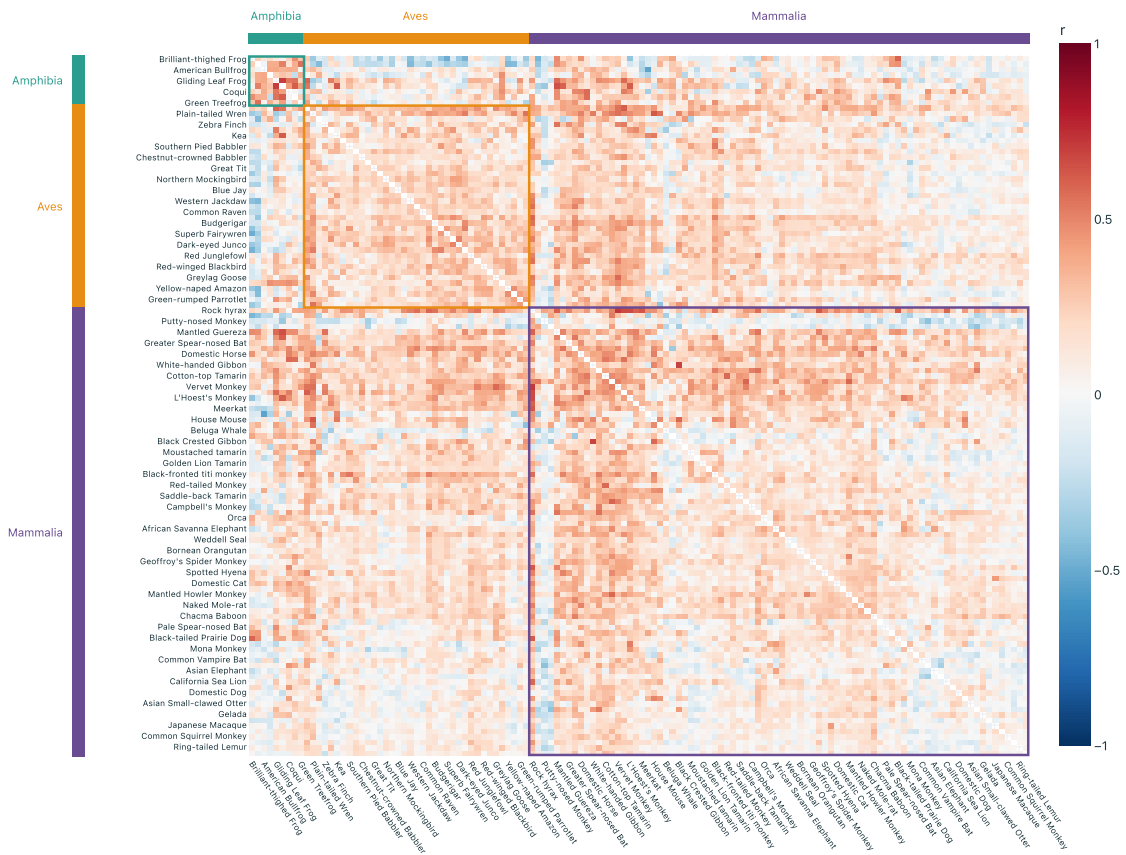


Figure 3. Species-pairwise acoustic-semantic correlation. Each cell shows the Pearson r between the vector of all acoustic-similarity values and the vector of all semantic-similarity values for every species pair. Species are ordered and grouped by taxonomic class as in the web application; hover-only pair counts from the interactive version are encoded in the same underlying figure data.

4 Form-to-meaning mapping: cracking the code?

To decompose the signal into interpretable units, we computed pointwise mutual information (PMI) between acoustic features and semantic keywords across all calls (Fig. 4). The strongest positive cell links `multi_component` with `combinatorial` ($\text{PMI} = 2.27$ bits), whereas the strongest depletion links `high_frequency` with `combinatorial` ($\text{PMI} = -3.70$ bits). The complete matrix is more informative than a small set of selected cells and is therefore reported using the same grouped visualization as the web application.

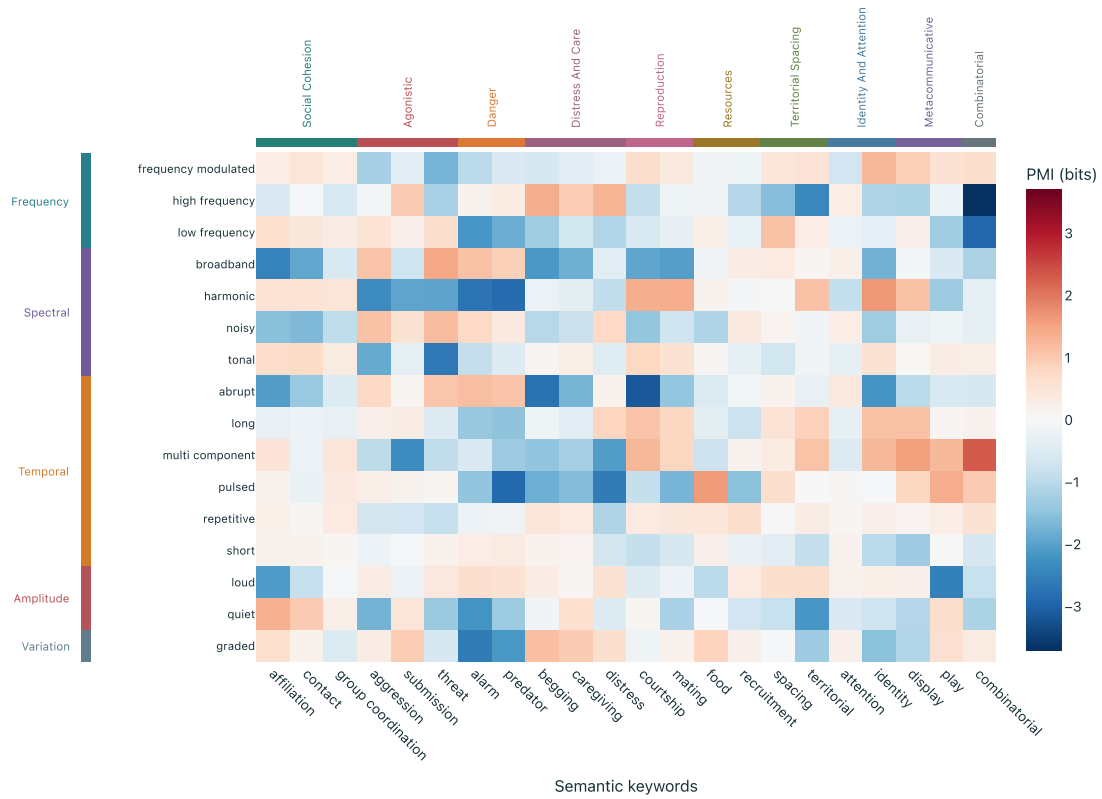


Figure 4. Keyword-level structure of the biological code. Pointwise mutual information (PMI, bits) between semantic functions (columns, grouped into panels by category) and acoustic features (rows, grouped by type), estimated across all 1,507 calls in AnimalLex. Rows and columns are ordered by the same acoustic and semantic groups used in the web application. Warm colors indicate over-representation and cool colors indicate under-representation; the interactive figure additionally exposes joint counts, shuffle expectations, permutation p values, and FDR-adjusted q values on hover.

5 Form-to-meaning mapping is learnable and generalizes across taxa

We trained models to predict semantic embeddings from acoustic embeddings under three hold-out conditions (Fig. 5). For held-out calls, ridge regression achieves cosine similarity 0.642 and MRR 0.126, compared with 0.343 and 0.036 for random pairing. Performance remains stable for species held out as complete groups (ridge cosine 0.638; MRR 0.125) and for held-out families (ridge cosine 0.633; MRR 0.111). Acoustic nearest-neighbour retrieval yields lower cosine similarity but higher MRR in every split, indicating that prediction and retrieval emphasize different aspects of the mapping.

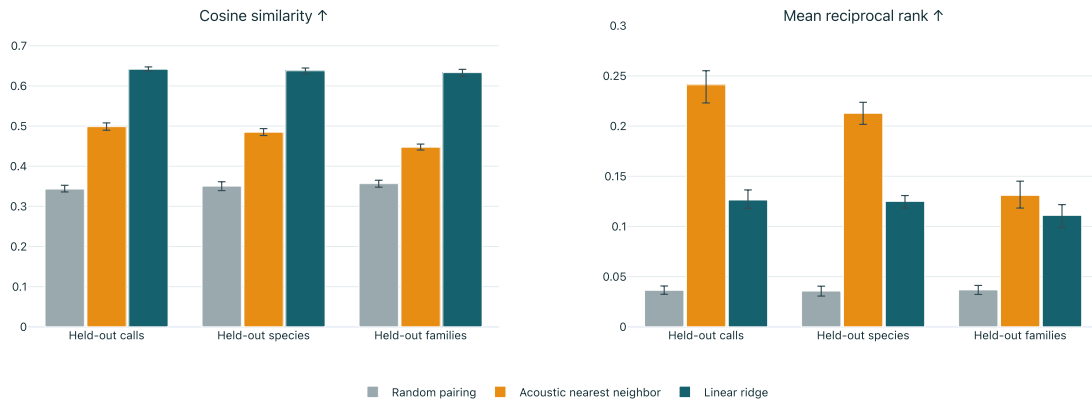


Figure 5. Cross-generalization of acoustic-to-semantic prediction. Three models are evaluated under three hold-out conditions of increasing difficulty. Cosine similarity measures agreement between predicted and true semantic embeddings; mean reciprocal rank (MRR) measures retrieval of the true call among candidates. The random baseline shuffles acoustic–semantic pairings within each test set. Acoustic nearest-neighbour retrieval returns the semantic embedding of the nearest training call. Bars and 95% bootstrap intervals are identical to the web application’s unfiltered prediction figure.

6 Lexical translation across species

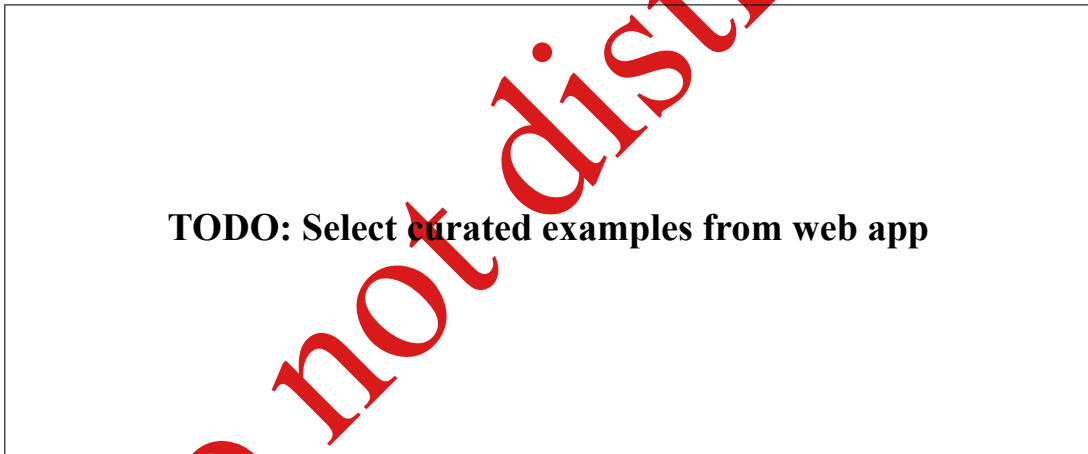


Figure 6. Selected cross-species acoustic–semantic motifs. The examples are manually selected from the web application’s complete results for their scientific or illustrative interest. The full, uncurated set of results is available in the web application.

AnimalLex allows best guess translations across the animal kingdom. Given a call from species A, we retrieve the call in species B whose semantic embedding is nearest—a lexical translation from one repertoire to another. When both species are represented in AnimalLex with full semantic annotations, the mapping is performed in semantic space by retrieval. Given the form-to-meaning structure identified above, translational attempts are also possible on the basis of acoustic annotations only—as will routinely be the case for poorly studied taxa—the acoustic-to-semantic predictor from the previous section supplies a best-guess semantic embedding from acoustic features alone, enabling translation even across the boundary of documented knowledge.

This two-mode pipeline (retrieval for annotated calls; prediction for unannotated ones) makes AnimalLex a living resource: every new species added expands the reach of translation, and the predictor provides a scaffold for annotating the next. The web application (<https://emmanuel-dev.tail3ea00b.ts.net/>) exposes the complete motif results and allows users to inspect correspondences interactively (see Appendix A). At the default acoustic and semantic thresholds of 0.65 and 0.65, respectively, the unfiltered analysis returns 73 motifs. Figure 6 is

intentionally reserved for a small manually selected subset whose provenance will remain the web application’s full result set.

A natural extension is to learn the translation *directly in acoustic space*, bypassing the semantic intermediate. Two species that are phylogenetically close should have nearly aligned acoustic spaces—similar vocal anatomies, shared evolutionary pressures, conserved call repertoires—and their repertoires should be mappable by a small rotation. Phylogenetically distant species may require a larger transformation. This suggests learning a family of species-pair rotations R_{AB} (Procrustes-style, constrained to be orthogonal) regularised by phylogenetic distance: $\|R_{AB} - \mathbf{I}\|_F \leq \lambda \cdot d_{\text{phylo}}(A, B)$, where d_{phylo} is branch length in a dated phylogeny and λ controls how much divergence is allowed per unit evolutionary distance. Such a model would be fully audio-grounded, require no semantic labels, and could scale to the thousands of species for which recordings but no annotations exist. We leave this to future work, but note that the form-to-meaning results above provide the theoretical justification: because acoustic and semantic spaces are aligned, a rotation in acoustic space is also, approximately, a rotation in semantic space.

7 Discussion

Our results establish three things. First, animal vocalizations carry systematic form-to-meaning relationships detectable across the tree of life—not just within well-studied model systems. Second, these relationships are learnable and generalizable: a model trained on some species predicts communicative meaning from acoustics for entirely held-out species and families. Third, this learnability enables practical cross-species lexical translation—finding, for any call, its closest semantic analogue in any other repertoire—including for species whose calls have never been semantically annotated.

Two non-exclusive mechanisms can explain the cross-taxon alignment. Convergent functional pressures may independently channel evolution toward similar acoustic solutions for similar communicative problems: the physics of sound localization favors broadband calls for alarm regardless of ancestry [Marler, 1955](#); the acoustics of long-range propagation favor tonality for cohesion calls [Morton, 1977](#). Phylogenetic inertia may additionally preserve ancestral form-to-meaning associations within lineages, consistent with the somewhat stronger same-family coefficient. These mechanisms predict different patterns in phylogenetically controlled analyses, which future work with broader taxonomic sampling can test.

Several caveats apply. AnimalLex represents the current state of synthesized knowledge, which skews toward well-studied mammals; birds and amphibians—acoustically diverse and experimentally tractable—are underrepresented. Our acoustic embeddings are derived from textual descriptions, not from audio directly; future work integrating audio representations [Hagiwara, 2022](#); [Robinson et al., 2025](#) will test whether the form-to-meaning alignment is stronger in the raw signal domain. Lexical translation should be understood as semantic proximity within the space of documented ethological descriptions—not as a claim about subjective experience, cognitive content, or the *Umwelt* of the signaling animal. Any mapping through human language inherits an additional anthropomorphic bias and should be treated as a heuristic rather than a literal gloss.

Despite these limitations, the demonstration that form-to-meaning structure is learnable and cross-taxon generalizable opens a practical path forward: with each new species added to AnimalLex, the translation model improves, and with each improvement the model can bootstrap annotation of the next species. This virtuous cycle—human knowledge, compressed by language models, refined by acoustic-semantic prediction, extended to undocumented taxa—is the long-term promise of the approach introduced here.

8 Methods

Database construction AnimalLex was assembled by prompting a large language model with a structured schema specifying six required fields per call: (1) call name, (2) free-text acoustic description (spectro-temporal features, amplitude envelope, call-internal structure), (3) free-text semantic description (communicative function, behavioral context, receiver responses), (4) ontology keywords drawn from a fixed controlled vocabulary, (5) subjective confidence (high, medium, or low), and (6) references. The present manuscript reports the web application’s *all calls* view: no call is excluded on the basis of confidence. Confidence-filtered views remain available in the application for sensitivity checks.

Embeddings Acoustic and semantic descriptions were embedded independently using the all-MiniLM-L6-v2 sentence transformer [Reimers and Gurevych, 2019](#), which maps variable-length text to a 384-dimensional vector space trained on semantic similarity tasks. Embeddings were ℓ_2 -normalized before computing cosine

distances. We deliberately used textual embeddings rather than audio-derived representations to ensure that acoustic and semantic descriptions were embedded in a comparable space; the limitations of this choice are discussed above.

Mantel test For each subset of calls (within species, same-family cross-species, and different-family pairs), we computed the Pearson correlation between acoustic and semantic distances. Statistical significance was assessed with the web application’s one-sided Mantel test using blocked label permutations: semantic call identities were shuffled within species. The shared bundle used for this manuscript and the app contains 999 permutations.

PMI analysis Acoustic and semantic keywords were taken from the structured annotation fields. Keywords occurring in fewer than four calls were excluded. For each acoustic–semantic keyword pair (a, s) , PMI was estimated as $\log_2 \hat{p}(a, s) - \log_2 \hat{p}(a) - \log_2 \hat{p}(s)$, where probabilities were estimated from call frequencies across the full database. Significance was assessed by shuffling complete acoustic-keyword sets among all calls while holding semantic-keyword sets fixed, using 999 permutations. We controlled the false-discovery rate across all tested cells using the Benjamini–Hochberg procedure.

Acoustic-to-semantic prediction We trained a ridge regression mapping acoustic embeddings $\mathbf{x} \in \mathbb{R}^{384}$ to semantic embeddings $\mathbf{y} \in \mathbb{R}^{384}$ and compared it with random pairing and acoustic nearest-neighbour retrieval. Predicted embeddings were ℓ_2 -normalized before evaluation. Performance was measured by (i) mean cosine similarity between predicted and true semantic embeddings and (ii) mean reciprocal rank (MRR) of the true call’s semantic embedding among candidates, ranked by cosine similarity to the prediction. Evaluation used 10 outer folds for held-out calls, species, and families; error bars are 95% percentile bootstrap intervals across outer folds.

Heavy AI methods This work made maximally heavy use of AI technology including database collection, curation and organization, discussion, design and implementation of analyses, literature search, writing and reviewing.

9 Contributions

Conceptualization: A.B., C.D. **Methodology:** A.B., C.D., E.F. **Software:** E.F., G.H. **Validation:** C.D., E.F. **Formal analysis:** A.B., E.F. **Investigation:** A.B., C.D. **Resources:** I.J. **Data curation:** A.B., C.D., E.F. **Writing – original draft:** A.B. **Writing – review & editing:** A.B., C.D., E.F., I.J. **Visualization:** E.F., G.H. **Supervision:** I.J. **Project administration:** I.J. **Funding acquisition:** I.J.

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A Web application

We release an interactive Dash web application at <https://emmanuel-dev.tail3ea00b.ts.net/>. It supports repertoire browsing; confidence-specific views; form-to-meaning, prediction, cross-species overlap, species-matrix, and keyword-association analyses; and exploration of acoustic–semantic motifs. The manuscript figures are static exports of the application’s plotting functions, and all reported manuscript values are generated from the same analysis bundle. Unless stated otherwise, both use the unfiltered *all calls* view.

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